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STGVAD: Spatio-Temporal Graph-based Vessel Behavior Anomaly Detection

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ABSTRACT Predicting vessel trajectories and detecting anomalous behaviors are fundamental yet challenging tasks in the maritime domain, primarily due to the dynamic and unstructured nature of the marine environment. While Spatio-Temporal Graph Neural Networks (ST-GNNs) have shown great promise in modeling relational and temporal dependencies, their reliance on fixed spatial anchors limits their applicability to maritime scenarios, where such anchors are inherently absent. To address this limitation, we propose a novel GNN-based framework, denoted as *Spatio-Temporal Graph-based Vessel behavior Anomaly Detection* (STGVAD), that redefines graph construction for vessel behavior modeling. Instead of relying on static spatial points, we represent each timestamped vessel state as a node and construct a unified multi-ship trajectory graph by linking temporally adjacent nodes and incorporating spatial proximity using the OPTICS clustering algorithm. This enables the joint modeling of temporal dynamics and inter-vessel interactions within a single graph structure. Our framework employs a GNN encoder to capture spatio-temporal patterns, followed by a time-series module. To support objective evaluation, we manually injected anomalies into real AIS dataset. Experimental results demonstrate that STGVAD significantly outperforms conventional time-series baselines, particularly under conditions with scarce anomalies. These findings highlight the importance of joint spatio-temporal modeling and validate the robustness and generalizability of our approach in complex maritime environments.

INDEX TERMS Vessel Behavior, Anomaly Detection, Graph Neural Networks (GNNs), Automatic identification system (AIS)

I. INTRODUCTION

Understanding and anticipating vessel behavior is a complex yet vital task within the maritime domain. The importance of this task spans across a wide range of applications including safety, surveillance, and traffic optimization [2], [28]. However, predicting vessel trajectories and identifying anomalous behaviors remain highly challenging due to the dynamic and unstructured nature of the marine environment.

Anomaly detection in vessel behavior is particularly difficult to define and quantify [4]. While numerous approaches

have been proposed in previous studies, most methods focus on individual vessel patterns, failing to account for interaction effects, especially in dense traffic situations where vessel proximity becomes a critical factor [30], [37]. This gap motivates the incorporation of graph-based representations to model inter-vessel relationships more effectively.

Graph Neural Networks (GNNs) offer a powerful solution for modeling such relational data. By aggregating information from neighboring nodes, GNNs inherently capture structural dependencies that are crucial for understanding interactions

among entities. GNNs have been successful in diverse domains including social networks [51], [57], traffic prediction [8], [19], and molecular dynamics [21], [29]. A particular subclass, Spatio-Temporal GNNs (ST-GNNs), has shown strong performance in modeling data with both spatial and temporal dimensions. These models typically utilize a fixed graph topology defined over static spatial points and apply sequential models like RNNs or LSTMs to encode temporal dynamics. However, in maritime environments, a very common assumption of traditional ST-GNNs often breaks down. Unlike road traffic or public transportation systems, where nodes typically represent fixed physical locations such as intersections or stations, the open sea lacks such clearly defined and static reference points. Vessels operate in a continuous and dynamic spatial domain where trajectories do not conform to predefined routes or stops. As a result, constructing a graph based on fixed spatial anchors becomes impractical. This hinders the direct application of conventional ST-GNN frameworks, which rely heavily on static graph topologies that are unsuitable for maritime settings.

Motivated by the core idea of conventional ST-GNNs, we propose a novel framework for vessel behavior anomaly detection, called *Spatio-Temporal Graph-based Vessel behavior Anomaly Detection (STGVAD)*. This framework preserves the spatio-temporal modeling capability through dedicated temporal and spatial modules, while introducing a fundamentally new graph construction strategy to address the unique challenges of maritime environments. Unlike conventional ST-GNNs that assume static spatial anchors, our framework models each timestamped vessel state as a graph node. By incorporating multiple time snapshots of vessel behavior into a single graph, we extend the graph into a multi-ship trajectory graph by including the ships that are within spatial proximity based on OPTICS algorithm. This formulation allows for a graph representation where both temporal and spatial dynamics are jointly encoded in a single coherent structure. STGVAD adopts a GNN-based encoder to process the constructed graphs and extract fine-grained spatio-temporal patterns, which are then fed into a time-series forecasting module to predict future vessel behavior and detect anomalies. In addition, to facilitate objective evaluation and address the lack of labeled maritime anomalies, we manually produced anomalous vessel trajectories based on real Automatic Identification System (AIS) dataset. This ensures a controlled yet realistic benchmark environment for assessing detection performance. We evaluate STGVAD against time-series models that disregard inter-vessel interactions, and demonstrate that our framework significantly outperforms these baselines, highlighting the importance of jointly modeling spatial and temporal dependencies. In addition, we compare STGVAD with existing vessel anomaly detection method and show that it achieves superior performance. To further validate the robustness of our framework, we evaluate STGVAD under diverse anomaly scenarios, including those with extremely scarce anomalous instances and show that it maintains strong performance even in subtle anomaly cases. This demonstrates

the framework's ability to generalize across varying degrees of anomaly ratio.

In summary, our main contributions are threefold:

- **Novel graph construction strategy:** We redefine the graph modeling process for maritime environments by treating each timestamped vessel state as a node and incorporating spatial proximity via the OPTICS clustering algorithm to form multi-ship trajectory graphs.
- **Joint spatio-temporal encoding:** Our framework simultaneously captures temporal dynamics and inter-vessel spatial interactions within a single graph structure, enabling fine-grained anomaly detection.
- **Controlled anomaly benchmark:** We manually curate anomalous vessel trajectories using real AIS data, providing a realistic yet controlled benchmark for maritime anomaly detection research.

II. RELATED WORKS

A. VESSEL BEHAVIOR ANOMALY DETECTION

1) Terminology

Before reviewing prior work, we clarify the terminology used in this paper. Vessel behavior refers to the set of motion patterns and operational states that can be derived from a ship's trajectory and onboard sensors.

Anomaly detection is widely defined as the problem of finding patterns in data that do not conform to expected behavior [5]. Recent work [33] further distinguishes between identified vessel behaviors or events and explicit anomalies in AIS streams. In line with the above definitions, we define vessel behavior anomaly detection as a supervised task that assigns each incoming AIS sequence to one of two classes: normal vessel behavior or anomalous vessel behavior. The following subsections review how this task has been approached to date and highlight the main open challenges.

2) Challenges

Vessel behavior anomaly detection faces two fundamental challenges: (1) the difficulty of defining anomalies with clear criteria, and (2) the scarcity of labeled anomalous data.

Vessel behavior anomalies lack universally consensus definitions. What constitutes abnormal behavior is highly context-sensitive [52]. It can vary significantly across geographic regions, traffic separation schemes, weather conditions, and vessel types. A trajectory that appears suspicious in one area may be considered entirely normal in another. This contextual variability makes it difficult to define fixed thresholds or to construct rule-based systems that generalize across maritime environments.

Also, labeled data for anomaly detection are extremely limited [13], [39]. Although the availability of high-quality labels can significantly improve model training and evaluation, annotating anomalies requires extensive domain knowledge and manual effort. This labeling process is often time-consuming, expensive, and difficult to scale. As a result, current research either focuses on detecting a single type of vessel behavior

in an unsupervised manner [22], [47], or proposes detecting predefined events using domain-specific behavior rules encoded as data queries [40]. While rule-based approaches provide interpretability, they often fail to adapt to unforeseen behaviors or evolving traffic patterns.

These challenges highlight the importance of establishing anomaly definitions and labeling criteria grounded in prior studies, while also underscoring the need for robust and adaptive models capable of detecting meaningful vessel behavior anomalies under diverse maritime conditions and limited labeled data.

B. DETECTION METHODS OF VESSEL ANOMALIES

Various approaches have been proposed to detect vessel anomalies, which can be broadly categorized into four types: (1) Rule-based and Statistical Methods, (2) Machine Learning-based Methods, (3) Deep Learning-based Methods, and (4) GNN-based Methods.

1) Rule-based and Statistical Methods

Rule-based approaches rely on predefined expert rules derived from COLREGs (International Regulations for Preventing Collisions at Sea) [26] or scenario-specific maritime knowledge [9]. Recent advancements including [1], [44] have incorporated ontology-based reasoning and semantic rule engines to enhance detection of abnormal vessel behaviors, such as deviations from expected routes or violations of restricted zones. These systems offer high interpretability and align well with regulatory standards. However, they are inherently limited by human-defined rules, making them susceptible to bias when faced with previously unseen or evolving maritime traffic patterns [43], [45].

To address these limitations, statistical methods have been introduced. Instead of relying on fixed rules, these methods model the probabilistic distributions of vessel behaviors and identify anomalies as significant deviations from expected patterns. Classical techniques include density-based clustering methods such as DBSCAN [17] and Gaussian Mixture Models [42]. More recent approaches employ stochastic frameworks such as the Ornstein–Uhlenbeck (OU) process [11] combined with multiple hypothesis testing to detect stealth behaviors and AIS spoofing without the need for predefined rules. These models automatically assess abnormality based on learned parameters and test statistics. While statistical models offer improved generalization compared to rule-based systems, they often struggle to capture complex dependencies in high-dimensional AIS data and may require dimensionality reduction or feature selection techniques to remain effective.

2) Machine Learning-based Methods

Machine learning-based methods have been introduced to overcome the rigidity of rule-based methods and the simplifying assumptions of statistical methods including linearity, independence, and stationarity.

Classification algorithms have been applied to detect abnormal vessel behaviors using AIS data [38], demonstrating improved adaptability over rule-based approaches. However, these methods require extensive labeled training data and struggle with previously unseen maritime traffic patterns. Clustering techniques have been employed for maritime anomaly detection [59], showing effectiveness in identifying suspicious vessel movements. Nevertheless, these approaches rely heavily on manual feature engineering and fail to capture complex temporal dynamics in vessel trajectories. Classification models for detecting anomalous maritime behaviors have been developed [46], achieving reasonable detection accuracy in controlled scenarios. Yet, these systems exhibit limited performance when dealing with inter-vessel dependencies and evolving maritime traffic conditions. Recent surveys [23] highlight the broader application of supervised and unsupervised models for trajectory prediction and maritime risk analysis, noting their reliance on historical AIS data. While these approaches offer better generalization than traditional methods, they often struggle to capture temporal dynamics and inter-vessel dependencies, limiting their performance in complex maritime scenarios where real-time adaptation and multi-vessel interaction modeling are crucial.

3) Deep Learning-based Methods

Deep learning-based methods have been introduced to overcome the limitations of machine learning-based approaches, such as the need for manual feature engineering and limited capacity to model complex spatio-temporal patterns.

These methods enable end-to-end learning directly from raw AIS data. For example, [41] has shown CNN-LSTM models to be effective in capturing both spatial and temporal dependencies for real-time anomaly detection. Attention-enhanced architectures like [6] further improve vessel classification performance by focusing on informative trajectory segments derived from multiple features. Other studies propose multi-task or fusion-based deep learning architectures to handle multiple maritime surveillance tasks simultaneously [10], [36], while case studies demonstrate the practical effectiveness of deep models in operational maritime settings [31], [55]. Despite these advances, a common limitation is that most deep learning models still underexplore inter-vessel interactions, which are critical for identifying relational or coordinated anomalies in dense maritime environments. Nevertheless, recent studies have shown steady improvements in performance. Representative models provide more refined representations of AIS data, enhance the accuracy of long-term prediction and anomaly detection, and also achieve progress in terms of computational efficiency.

GeoTrackNet [37] addresses anomaly detection by learning probabilistic neural representations of AIS tracks and applying an *a contrario* test to detect statistically unlikely deviations. It achieves scalable unsupervised detection but remains highly dependent on local traffic distributions, limiting cross-region generalization and overlooking coordinated behaviors. TrAISformer [35] represents the state of the art in trajectory

forecasting, introducing a sparse augmented AIS representation and a heterogeneity-aware loss within a Transformer backbone. It delivers strong long-range prediction accuracy, but its complexity poses challenges for real-time and large-scale deployment.

4) GNN-based Methods

Graph-based methods have recently been adopted for vessel trajectory prediction and anomaly detection, aiming to leverage the relational structure among maritime entities. However, many of these approaches suffer from two core limitations: the use of unnatural or overly abstract graph constructions and the superficial application of GNNs without fully exploiting their relational reasoning capabilities.

Several studies attempt to model vessel movement or behavior using spatio-temporal graphs. For instance, trajectory prediction models [50] typically rely on attention-based adjacency matrices built from LSTM-encoded motion features, but such graphs often lack physical meaning, failing to account for real-world navigational constraints such as proximity, relative bearing, or traffic rules. In practice, the resulting adjacency structure is learned indirectly in a latent feature space, rather than being derived from explicit and observable vessel interactions. This limits the physical interpretability of the graph and reduces consistency across diverse maritime scenarios.

A similar issue appears in anomaly detection models [27], [58], where graph attention networks (GATs) are applied over either temporal sequences or feature spaces. These graphs are commonly constructed as fully connected or sliding-window-based structures, assuming uniform interactions between timestamps or feature dimensions. As a result, the GNN acts more as an attention-weighted MLP than a relational message-passing model, and the network fails to capture higher-order or inter-vessel dependencies that are critical in dense maritime traffic scenarios.

Some approaches [12] formulate AIS data as port-to-port movement graphs over discrete time intervals. While these methods do introduce structured connectivity, they reduce vessel dynamics to coarse-grained origin-destination transitions, discarding valuable intra-trajectory behavior such as acceleration, course change, or anomalous detours. Moreover, their GNN components often operate in a static, per-snapshot fashion, missing the temporal evolution of graph topology.

In summary, existing GNN-based methods often fail to fully leverage the structural expressiveness of graph learning. Meanwhile, in other domains such as traffic forecasting [24], [53], [54], pedestrian trajectory prediction [34], and real-time incident detection [14], ST-GNNs have achieved notable success by integrating spatial aggregation with temporal encoders.

Despite these advancements, most ST-GNN models rely on the assumption of fixed node anchors such as predefined locations or persistent entity identifiers, which allows for a static graph topology to be maintained across time steps. However, this assumption is incompatible with the maritime

domain, where vessels operate in a fluid and dynamic environment. Ships frequently change their positions, enter or leave monitored regions, and interact across highly variable spatial and temporal scales.

Existing adaptations of ST-GNNs to maritime data often simplify vessel dynamics through static or snapshot-based modeling pipelines. These approaches tend to overlook temporally evolving interactions and fine-grained behavioral patterns, limiting their ability to generalize across diverse maritime scenarios.

While existing studies have advanced through rule-based, machine learning, deep learning, and graph-based methods, they still fall short of fully capturing the complexity of maritime environments and the intricate inter-vessel interactions essential for anomaly detection. These limitations highlight the need for novel approaches that can effectively model both temporal dynamics and spatial relationships in maritime settings. To address these challenges, we propose STGVAD. In this framework, each timestamped vessel state is modeled as a node, and edges are dynamically constructed based on temporal adjacency and spatial proximity. This design enables the formation of semantically consistent interaction graphs, allowing the model to simultaneously capture temporal dynamics and inter-vessel dependencies in complex maritime environments.

III. METHODOLOGY

Figure 1 presents an overview of STGVAD, our proposed anomaly detection framework. The model processes AIS trajectory data by constructing spatio-temporal graphs that reflect both intra- and inter-vessel relationships. These graphs are then used to learn node representations through a combination of GNNs and temporal models, ultimately enabling the detection of anomalous behavior in vessel trajectories.

A. PROBLEM DEFINITION

Given AIS trajectory data $\mathcal{X} = \{\mathbf{x}_i^t | i \in \mathcal{V}, t \in T\}$, where $\mathbf{x}_i^t$ represents the feature vector of vessel i at timestamp t containing maritime information which are latitude, longitude, speed over ground (SOG), and course over ground (COG), our objective is to perform binary classification to determine whether individual vessel trajectories exhibit anomalous behavior. For a trajectory window of size w , we consider a sequence of observations $\{\mathbf{x}_i^t, \mathbf{x}_i^{t+1}, \dots, \mathbf{x}_i^{t+w-1}\}$ spanning from timestamp t to $t + w - 1$, and aim to classify this entire trajectory segment as either normal or anomalous based on the spatio-temporal patterns encoded within the sequence. The problem can be formulated as learning a function $f : \mathbb{R}^{w \times d} \rightarrow \{0, 1\}$, where d is the feature dimension and the output represents the binary anomaly classification.

B. GRAPH CONSTRUCTION

In order to leverage GNNs for the purpose of learning vessel trajectory representations and updating node embeddings through message passing, STGVAD constructs spatio-temporal graphs that capture both individual vessel dynamics

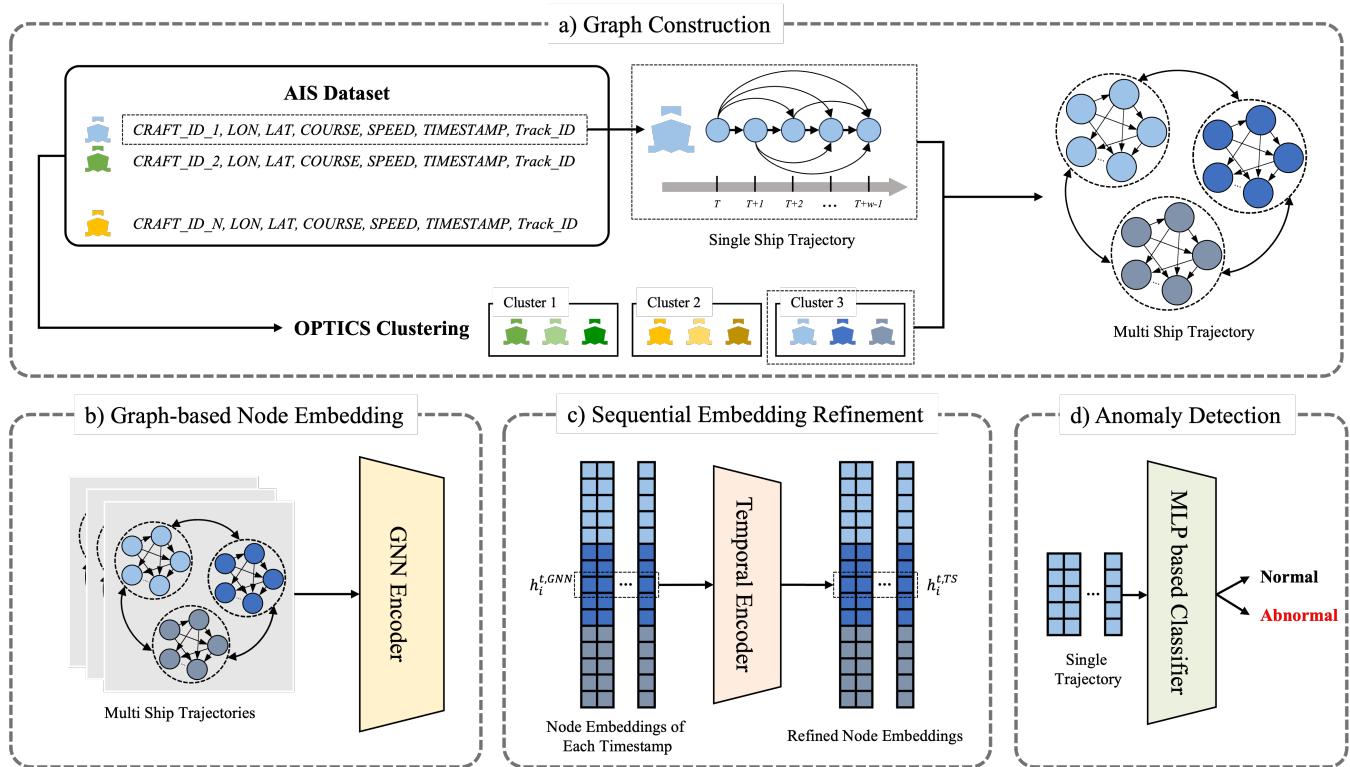


FIGURE 1: Overview of STGVAD, the proposed anomaly detection framework. The proposed framework processes AIS trajectory data by constructing spatio-temporal graphs that encapsulate both temporal connections within individual vessels and spatial interactions between different vessels. GNNs update the node representations based on these graphs, and a subsequent temporal module enhances the embeddings to account for long-range temporal dependencies. The resulting trajectory-level features are then utilized to perform binary classification for anomaly detection.

and inter-vessel interactions. Conventional ST-GNN frameworks model nodes as immovable spatial anchors, thereby enabling the precomputation of the underlying adjacency matrix, which can be reused at every time step [14], [25], [34].

In maritime environments, however, the spatial relationships among vessels change continuously during navigation. Ships that are spatially separated at time t may converge to close proximity at time $t + 1$, then diverge again at subsequent timestamps. It is evident that a static adjacency matrix cannot capture these dynamic inter-vessel relationships, which are fundamental to modeling maritime traffic patterns and safety-critical interactions.

To address this challenge, we construct dynamic spatio-temporal graphs where adjacency relationships are recomputed at each timestamp based on actual vessel positions and trajectory window. We cluster co-located vessels using OPTICS and create inter-vessel edges that are bidirectional when vessels share the same timestamp, and directed from earlier to later timestamps otherwise. This approach yields multi-ship graphs that integrate individual vessel motion with context-aware inter-vessel interactions.

1) Single-ship Trajectory Graph

The foundation of our graph construction is predicated on the modeling of individual vessel trajectories as spatio-temporal graphs. For each vessel, a set of nodes is created to represent its state at consecutive timestamps within a sliding window of size w . Directed edges are established from each timestamp to all subsequent timestamps within the specified window, as formalized in Algorithm 1. This procedure is applied independently to each vessel in the dataset to construct a corresponding single-ship trajectory graph. This is expressed as $t \rightarrow t + k$ for $k = 1, 2, \dots, w - 1$. The establishment of directed edges facilitates the capture of both short-term and long-term temporal dependencies in the trajectory. This results in the formation of a graph for each individual vessel, denoted by $\mathcal{G}_i = (\mathcal{V}_i, \mathcal{E}_i)$, where $\mathcal{V}_i$ denotes the timestamp nodes and $\mathcal{E}_i$ represents the temporal edges. This approach demonstrates the efficacy of capturing temporal dependencies within a graph structure, thereby enabling GNNs to update node embeddings through the propagation of information along these temporal edges, a process that diverges from the reliance on traditional time-series prediction models such as LSTMs or Transformers.

Algorithm 1 Single-ship Trajectory Graph Construction

Input: Ordered timestamps $t_1, t_2, \dots, t_n$ of a fixed vessel; window size w

Output: Single-ship trajectory graphs $\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_m$

```

1: for each sliding window  $w_k = [t_k, t_{k+1}, \dots, t_{k+w-1}]$  do
2:   Initialize graph  $\mathcal{G}_k = (\mathcal{V}_k, \mathcal{E}_k)$ 
3:    $\mathcal{V}_k \leftarrow \{v_t \mid t \in w_k\}$ 
4:   for each pair  $(t_1, t_2)$  such that  $t_1, t_2 \in w_k$  and  $t_1 < t_2$ 
       do
5:     Add directed edge  $(v_{t_1} \rightarrow v_{t_2})$  to  $\mathcal{E}_k$ 
6:   end for
7: end for
8: return  $\{\mathcal{G}_k\}_{k=1}^m$ 

```

2) Multi-ship Trajectory Graph

To construct multi-ship spatio-temporal graphs under the evolving maritime context, STGVAD leverages the OPTICS clustering algorithm [3] to group vessels based on their geographical positions at a specific timestamp t . The primary goal of this clustering step is to identify vessels in close spatial proximity at that moment, under the assumption that vessels co-located at t will likely remain relatively close for a certain period, thereby sharing similar navigational contexts and potential interactions.

Building upon this assumption, we construct multi-ship trajectory graphs by using the vessels identified within each cluster at t as anchors for graph expansion. For each vessel in the cluster, we generate a single-ship trajectory graph starting from t , capturing its temporal evolution across subsequent timestamps within a sliding window. These individual trajectory graphs are then interlinked within the cluster to capture inter-vessel interactions over time.

Specifically, for vessels i and j within the same cluster at t , we establish edges to encode their temporal relationships: if i and j are co-located at the exact same timestamp, we create a bidirectional edge; if their observations occur at different timestamps, we create a directed edge from the earlier to the later timestamp. This design incorporates spatial proximity from clustering and temporal ordering from trajectory expansion. This allows the GNN to propagate information across both dimensions when updating node representations.

Overall, this construction unifies the temporal dependencies within individual vessel trajectories and the spatial interactions among proximate vessels, providing a structured and context-aware foundation for GNN-based learning of collective maritime behavior patterns.

The detailed construction procedure for the multi-ship trajectory graph is summarized in Algorithm 2.

C. NODE EMBEDDING

The node embedding approach in STGVAD draws inspiration from the two-stage update mechanism that is commonly employed in ST-GNN frameworks [54]. The objective of adapting this mechanism is to capture both spatial and tempo-

Algorithm 2 Multi-ship Trajectory Graph Construction

Input: All vessel positions at timestamp t ; window size w

Output: Multi-ship trajectory graph $\mathcal{G}_t$

```

1: Apply OPTICS clustering on vessel positions at  $t$  to
   obtain clusters  $\{\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_c\}$ 
2: for each cluster  $\mathcal{C}_i$  do
3:   Initialize graph  $\mathcal{G}_t^{(i)} = (\mathcal{V}_t^{(i)}, \mathcal{E}_t^{(i)})$ 
4:   for each vessel  $v \in \mathcal{C}_i$  do
5:     Construct single-ship trajectory graph  $\mathcal{G}_v$  using Al-
       gorithm 1 with anchor at  $t$ 
6:     Add nodes and edges from  $\mathcal{G}_v$  to  $\mathcal{G}_t^{(i)}$ 
7:   end for
8:   for each pair of vessels  $(v_i, v_j) \in \mathcal{C}_i$  do
9:     for each pair of timestamps  $(t_p, t_q)$  within the win-
       dow of size  $w$  do
10:      if  $t_p = t_q$  then
11:        Add bidirectional edge between  $v_i^{t_p}$  and  $v_j^{t_q}$  to
           $\mathcal{E}_t^{(i)}$ 
12:      else if  $t_p < t_q$  then
13:        Add directed edge  $v_i^{t_p} \rightarrow v_j^{t_q}$  to  $\mathcal{E}_t^{(i)}$ 
14:      end if
15:    end for
16:   end for
17: end for
18: return  $\{\mathcal{G}_t^{(i)}\}_{i=1}^c$ 

```

ral dependencies in maritime trajectory data. The embedding process is comprised of two sequential stages. First, spatio-temporal embedding updates are performed through graph neural networks. Second, temporal sequence refinement is conducted using time-series models.

1) Graph-based Embedding Update

In the first stage, we update node embeddings by leveraging both spatial and temporal relationships captured in the constructed spatio-temporal graph. We employ established GNN architectures including GCN, GAT, and GraphSAGE to aggregate information from neighboring nodes. For a given node v_i^t representing vessel i at timestamp t , the spatio-temporal embedding update can be formulated as:

$$\mathbf{h}_i^{t, \text{GNN}} = \text{GNN}(\mathbf{x}_i^t, \{\mathbf{x}_j^s \mid (j, s) \in \mathcal{N}_i^t\})$$

Here, $\mathbf{h}_i^{t, \text{GNN}}$ denotes the updated embedding, $\mathbf{x}_i^t$ is the initial node feature, and $\mathcal{N}_i^t$ represents the neighborhood of vessel i at timestamp t . The neighborhood $\mathcal{N}_i^t$ includes two types of nodes: temporal neighbors, which correspond to the same vessel i at different timestamps (e.g., v_i^{t-1}, v_i^{t+1}), and spatial neighbors, which correspond to different vessels that co-occur with vessel i at timestamp t (e.g., v_j^t where $j \neq i$). This stage enables the model to incorporate both contextual information from spatially proximate vessels and temporal dependencies from the vessel's own trajectory.

2) Sequential Embedding Refinement

In the second stage, we further refine the node embeddings by capturing long-term temporal dependencies using time-series models such as LSTM networks or Transformer architectures. The graph-based updated embeddings from the previous stage serve as input to the temporal model:

$$\begin{aligned} \mathbf{h}_i^{t,TS} &= f_{\text{temporal}}(\mathbf{h}_i^{t,GNN}) \\ &\vdots \\ \mathbf{h}_i^{t+w-1,TS} &= f_{\text{temporal}}(\mathbf{h}_i^{t,GNN}, \mathbf{h}_i^{t+1,GNN}, \dots, \mathbf{h}_i^{t+w-1,GNN}) \end{aligned}$$

where f_{temporal} represents the temporal modeling function, and $\mathbf{h}_i^{t,TS}$ represents the final node embedding after sequential refinement. This two-stage approach allows the model to effectively capture the complex spatio-temporal dynamics inherent in maritime vessel trajectories, with the first stage handling local spatio-temporal patterns and the second stage modeling sequential temporal evolution.

D. ANOMALY DETECTION

After obtaining the refined node embeddings from the two-stage spatio-temporal learning process, STGVAD performs anomaly detection at the individual vessel trajectory level. The final task involves classifying whether each vessel's trajectory within a given window exhibits anomalous behavior, formulated as a binary classification problem.

For each vessel i with a trajectory window of size w , we collect the sequence of refined embeddings $\{\mathbf{h}_i^{t,TS}, \mathbf{h}_i^{t+1,TS}, \dots, \mathbf{h}_i^{t+w-1,TS}\}$ where each embedding $\mathbf{h}_i^{t,TS} \in \mathbb{R}^d$ represents the spatio-temporally learned representation at timestamp t . These embeddings are concatenated to form a trajectory-level representation $\mathbf{H}_i \in \mathbb{R}^{w \times d}$ that captures the collective behavior patterns within the window.

The anomaly detection function can be formulated as:

$$y_i = f_{\text{classifier}}(\mathbf{H}_i) : \mathbb{R}^{w \times d} \rightarrow \{0, 1\}$$

where $f_{\text{classifier}}$ is a classification model that maps the trajectory representation to a binary output, with $y_i = 1$ indicating an anomalous trajectory and $y_i = 0$ representing normal behavior. This vessel-level classification approach enables the identification of irregular maritime activities by analyzing the learned spatio-temporal patterns encoded in the trajectory window rather than detecting point-wise anomalies at individual timestamps.

IV. EXPERIMENT

A. EXPERIMENTAL SETUP

1) Dataset

Figure 2 illustrates the AIS trajectories used in our experiments, extracted from the OMTAD dataset [32], an in-house re-export of AIS logs in the West Pacific grid for the years 2018–2020. The chosen West-Pacific region corresponds to dense, heavily trafficked shipping lanes off Australia's west coast. Each record includes the vessel identifier (CRAFT ID), geolocation (LAT, LON), kinematic information such as COG

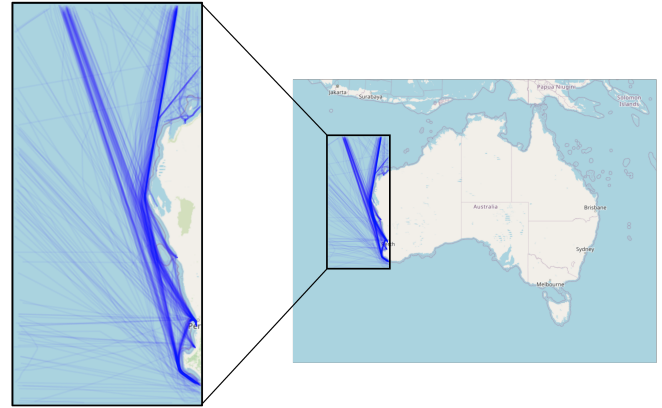


FIGURE 2: AIS vessel trajectories from the OMTAD dataset in the West-Pacific region off the western coast of Australia. Trajectories are visualized using semi-transparent blue lines, with darker paths indicating frequently traveled shipping routes.

and SOG, a UTC timestamp rounded to the nearest hour, and a system-generated track ID that represents individual vessel trajectories.

Data Preprocessing

The transformation of angular features, such as COG, into sine and cosine components is a well-established technique in machine learning, particularly when dealing with cyclical data. This approach addresses the discontinuity at the $0^\circ/360^\circ$ boundary, ensuring smooth transitions and preserving the inherent periodicity of the data. This transformation ensures that the model receives continuous and bounded inputs, facilitating better learning and generalization.

Next, for each track ID, positions are resampled to a strict one-hour grid and linearly interpolated where gaps exist. Tracks shorter than 10 hours are discarded, thus producing length-balanced sequences and avoiding sparsity.

Graph Preparation

As illustrated in Figure 3, which shows the AIS vessel positions at a specific timestamp and the clustering results, we construct interaction graphs based on the spatial distribution of vessels captured in the OMTAD dataset. At each timestamp t , we apply the OPTICS clustering algorithm [3] to these spatial snapshots, grouping spatially proximate vessels into dynamic clusters without imposing predefined constraints on the number or shape of clusters. This approach enables us to capture the fluid and irregular movement patterns inherent in maritime traffic while providing a structured foundation for subsequent graph-based modeling.

To facilitate a transductive GNN setting where each test node shares a fixed graph context with training nodes, we sample a fixed number k of vessel trajectories from each cluster. By keeping k constant, the adjacency matrix size remains the same across all graphs. For each sampled set, we assemble a directed temporal graph over a w hour observation

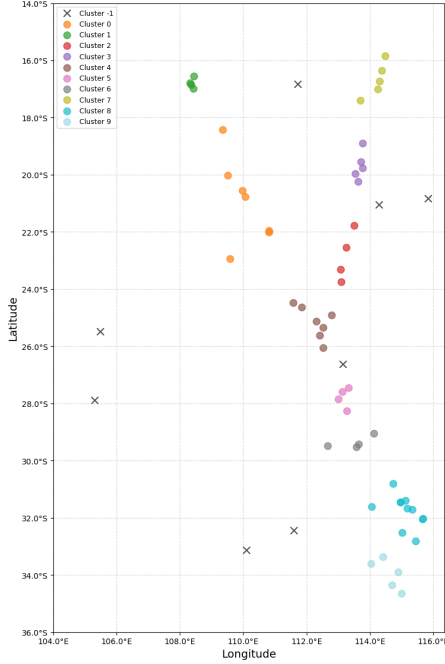


FIGURE 3: OPTICS clustering results on AIS vessel positions from the OMTAD dataset in the West-Pacific region at 14:00 UTC on April 19, 2018. Each point represents a vessel's position at this timestamp, colored according to its assigned cluster. Black 'x' markers denote noise points identified by the algorithm, while colored circular markers indicate clustered vessel positions.

window: each node corresponds to one time-stamped vessel state, resulting in exactly $k \times w$ nodes per graph.

Anomaly Injection

To address the lack of ground truth labels in maritime data, we synthesize anomalies by injecting artificial perturbations into normal trajectory data. Anomaly injection is applied at each node of the specified trajectories. The proportion of anomalous trajectories in this injection is determined by the specified trajectory anomaly ratio r_{traj} . This procedure is designed to emulate realistic fault patterns, where anomalous nodes appear sequentially rather than in an i.i.d. fashion.

Let w be the length of a trajectory and $r_{node} \in \{r_1, r_2, r_3\}$ denote the node anomaly ratio chosen at node level. We first define an anomaly block of length $m = r_{node}w$ and sample a random start index $s \sim \mathcal{U}(0, w - m)$. This defines a binary anomaly mask z_t for the trajectory:

$$z_t = \begin{cases} 1, & \text{if } s \leq t < s + m, \\ 0, & \text{otherwise,} \end{cases} \quad t = 0, \dots, w - 1.$$

For the nodes marked as anomalous ($z_t = 1$), we inject perturbations into their Speed Over Ground (SOG) and Course Over Ground (COG) features [41].

Specifically, we first model the statistical properties of normal vessel movements. We assume that the rate of change in SOG (a) and COG (ω) follows a normal distribution:

$$a \sim \mathcal{N}(\mu_a, \sigma_a^2), \quad \omega \sim \mathcal{N}(\mu_\omega, \sigma_\omega^2),$$

where

$$a_i = \frac{\text{SOG}_i - \text{SOG}_{i-1}}{\Delta t}, \quad \omega_i = \frac{\text{COG}_i - \text{COG}_{i-1}}{\Delta t}.$$

To introduce a significant deviation, we generate anomalous rates a_i^* and ω_i^* using a scaling factor $k > 3$:

$$a_i^* = \mu_a + k \cdot \sigma_a, \quad \omega_i^* = \mu_\omega + k \cdot \sigma_\omega.$$

The choice of $k > 3$ ensures the perturbed values lie outside the 99.7% confidence interval of the normal distribution. The corresponding SOG and COG values are then updated iteratively for the duration of the anomaly block:

$$\text{SOG}_i^* = \text{SOG}_{i-1} + a_i^* \cdot \Delta t,$$

$$\text{COG}_i^* = \text{COG}_{i-1} + \omega_i^* \cdot \Delta t.$$

Finally, an entire trajectory is labeled as an anomalous trajectory ($y_{traj} = 1$) if it contains at least one perturbed node ($\sum_{t=0}^{w-1} z_t > 0$), and as normal ($y_{traj} = 0$) otherwise. This approach allows us to generate realistic and challenging anomalies for training and evaluating our detection models.

2) Hardware Setup

All experiments are conducted on four NVIDIA RTX 3090 GPUs (24 GB each) within a Docker container (Ubuntu 20.04, CUDA 12.3, PyTorch 2.2).

3) Baselines

To evaluate the effectiveness of our framework, we compare it against purely time-series-based models that do not account for inter-ship relationships. All models discussed in this section operate according to the following common pipeline. Clustered and temporally sliced track sequences are fed into the model, which transforms them into embedding representations. The embeddings are projected via a linear layer to produce logits, which are then passed through a sigmoid classifier to yield per-track anomaly probabilities.

- **LSTM [18]**: It captures temporal dependencies in each track sequence using a recurrent architecture. The final hidden state summarizes the track's temporal dynamics and is used to predict its anomaly score.
- **Transformer [48]**: It models the temporal relationships within tracks using self-attention. The output embeddings of all time steps obtain a track-level representation, which is then used for anomaly prediction.
- **TrAISformer [35]**: It represents one of the most advanced approaches for AIS data embedding, addressing multimodal vessel trajectories through sparse high-dimensional representation and classification-based loss functions. We adapt this state-of-the-art model for binary anomaly classification to serve as a baseline in our trajectory-level anomaly detection experiments.

TABLE 1: Performance of time-series-based models and spatial GNN-integrated variants under different trajectory anomaly ratios (r_{traj}). Node anomaly ratio is fixed at $r_{node} = 0.5$. Values are reported as mean $\pm$ standard deviation. The best result per column is highlighted in **bold**, and the second-best is underlined. Here, TRANS denotes the Transformer model.

(a) Low Trajectory Anomaly Ratio ($r_{traj} = 0.1$)

Method		Prec	Rec	F ₁	Acc	PR-AUC	G-mean	MCC
Time-series	LSTM	83.61 $\pm$ 1.02	79.41 $\pm$ 1.27	81.52 $\pm$ 1.06	96.44 $\pm$ 0.34	91.15 $\pm$ 1.28	91.26 $\pm$ 0.38	79.48 $\pm$ 1.60
	TRANS	84.96 $\pm$ 1.23	73.74 $\pm$ 1.87	78.98 $\pm$ 1.29	96.02 $\pm$ 0.40	90.81 $\pm$ 1.21	89.78 $\pm$ 0.35	77.00 $\pm$ 1.67
	TrAISformer	88.89 $\pm$ 1.29	84.45 $\pm$ 1.35	86.60 $\pm$ 0.37	97.38 $\pm$ 0.06	93.28 $\pm$ 0.42	91.35 $\pm$ 0.66	85.19 $\pm$ 0.39
STGVAD (Ours)	LSTM+GCN	88.14 $\pm$ 0.92	78.20 $\pm$ 0.98	82.93 $\pm$ 0.67	96.82 $\pm$ 0.41	92.12 $\pm$ 1.32	91.64 $\pm$ 0.38	86.14 $\pm$ 0.77
	LSTM+SAGE	91.45 $\pm$ 0.68	85.83 $\pm$ 0.47	88.52 $\pm$ 0.45	97.84 $\pm$ 0.33	95.78 $\pm$ 0.85	92.12 $\pm$ 0.35	87.33 $\pm$ 0.39
	LSTM+GAT	94.43 $\pm$ 0.70	87.45 $\pm$ 0.58	89.48 $\pm$ 0.68	98.20 $\pm$ 0.22	93.23 $\pm$ 1.60	93.24 $\pm$ 0.32	89.86 $\pm$ 0.47
	TRANS+GCN	83.79 $\pm$ 1.21	78.13 $\pm$ 0.13	80.97 $\pm$ 0.02	96.30 $\pm$ 0.27	93.18 $\pm$ 0.07	91.88 $\pm$ 0.49	78.86 $\pm$ 0.32
	TRANS+SAGE	84.80 $\pm$ 1.19	77.95 $\pm$ 0.40	81.20 $\pm$ 0.31	96.42 $\pm$ 0.26	<u>94.13</u> $\pm$ 0.40	90.04 $\pm$ 0.40	79.27 $\pm$ 0.19
	TRANS+GAT	84.41 $\pm$ 1.12	68.20 $\pm$ 0.47	75.50 $\pm$ 0.20	95.57 $\pm$ 0.24	91.27 $\pm$ 0.43	91.34 $\pm$ 0.47	73.54 $\pm$ 0.42

(b) High Trajectory Anomaly Ratio ($r_{traj} = 0.3$)

Method		Prec	Rec	F ₁	Acc	PR-AUC	G-mean	MCC
Time-series	LSTM	90.59 $\pm$ 0.45	88.42 $\pm$ 0.97	89.54 $\pm$ 0.92	93.77 $\pm$ 0.29	96.49 $\pm$ 0.13	93.83 $\pm$ 0.29	85.04 $\pm$ 0.99
	TRANS	91.02 $\pm$ 0.60	85.97 $\pm$ 0.69	88.47 $\pm$ 0.79	93.24 $\pm$ 0.29	96.31 $\pm$ 0.45	91.78 $\pm$ 0.35	83.63 $\pm$ 0.99
	TrAISformer	94.32 $\pm$ 0.26	94.27 $\pm$ 0.48	94.29 $\pm$ 0.27	96.57 $\pm$ 0.14	98.64 $\pm$ 0.10	95.90 $\pm$ 0.24	91.85 $\pm$ 0.37
STGVAD (Ours)	LSTM+GCN	94.36 $\pm$ 0.35	90.51 $\pm$ 0.92	92.41 $\pm$ 0.66	95.51 $\pm$ 0.25	99.02 $\pm$ 0.15	94.98 $\pm$ 0.29	91.84 $\pm$ 0.84
	LSTM+SAGE	94.78 $\pm$ 0.50	<u>95.10</u> $\pm$ 0.30	<u>94.91</u> $\pm$ 0.29	<u>96.90</u> $\pm$ 0.16	99.34 $\pm$ 0.11	<u>96.57</u> $\pm$ 0.32	<u>92.75</u> $\pm$ 0.41
	LSTM+GAT	96.88 $\pm$ 0.41	96.00 $\pm$ 0.30	96.58 $\pm$ 0.45	97.93 $\pm$ 0.20	98.91 $\pm$ 0.33	97.11 $\pm$ 0.35	94.95 $\pm$ 0.58
	TRANS+GCN	<u>95.69</u> $\pm$ 0.73	92.24 $\pm$ 0.39	93.97 $\pm$ 0.04	96.43 $\pm$ 0.16	<u>99.17</u> $\pm$ 0.05	95.27 $\pm$ 0.22	91.41 $\pm$ 0.29
	TRANS+SAGE	89.01 $\pm$ 0.52	89.69 $\pm$ 0.35	89.34 $\pm$ 0.27	93.56 $\pm$ 0.17	98.46 $\pm$ 0.27	93.04 $\pm$ 0.40	84.68 $\pm$ 0.29
	TRANS+GAT	91.69 $\pm$ 0.30	87.76 $\pm$ 0.34	89.72 $\pm$ 0.16	93.98 $\pm$ 0.18	<u>99.17</u> $\pm$ 0.15	92.54 $\pm$ 0.47	85.40 $\pm$ 0.27

(c) Stress Trajectory Anomaly Ratio ($r_{traj} = 0.5$)

Method		Prec	Rec	F ₁	Acc	PR-AUC	G-mean	MCC
Time-series	LSTM	93.25 $\pm$ 0.51	93.20 $\pm$ 0.49	93.30 $\pm$ 0.45	93.24 $\pm$ 0.45	97.78 $\pm$ 0.05	94.83 $\pm$ 0.22	86.42 $\pm$ 0.90
	TRANS	92.94 $\pm$ 0.52	92.55 $\pm$ 0.45	92.76 $\pm$ 0.45	92.72 $\pm$ 0.45	98.11 $\pm$ 0.21	95.19 $\pm$ 0.22	85.40 $\pm$ 0.89
	TrAISformer	96.10 $\pm$ 0.24	96.25 $\pm$ 0.28	96.17 $\pm$ 0.17	96.16 $\pm$ 0.17	99.39 $\pm$ 0.03	96.16 $\pm$ 0.17	92.32 $\pm$ 0.34
STGVAD (Ours)	LSTM+GCN	93.66 $\pm$ 0.20	94.94 $\pm$ 0.44	94.30 $\pm$ 0.44	94.21 $\pm$ 0.45	<u>99.61</u> $\pm$ 0.19	96.82 $\pm$ 0.25	92.11 $\pm$ 0.89
	LSTM+SAGE	96.90 $\pm$ 0.54	97.42 $\pm$ 0.40	97.20 $\pm$ 0.40	97.12 $\pm$ 0.30	99.73 $\pm$ 0.07	98.24 $\pm$ 0.21	94.29 $\pm$ 0.20
	LSTM+GAT	<u>96.85</u> $\pm$ 0.32	<u>97.39</u> $\pm$ 0.50	<u>97.10</u> $\pm$ 0.29	<u>97.10</u> $\pm$ 0.45	99.60 $\pm$ 0.03	<u>98.19</u> $\pm$ 0.22	<u>94.11</u> $\pm$ 0.40
	TRANS+GCN	95.46 $\pm$ 0.45	94.70 $\pm$ 0.40	95.11 $\pm$ 0.09	95.18 $\pm$ 0.20	99.60 $\pm$ 0.04	96.78 $\pm$ 0.25	90.13 $\pm$ 0.16
	TRANS+SAGE	93.67 $\pm$ 0.71	93.48 $\pm$ 0.38	93.65 $\pm$ 0.39	93.50 $\pm$ 0.18	99.16 $\pm$ 0.08	95.98 $\pm$ 0.25	87.04 $\pm$ 0.16
	TRANS+GAT	93.42 $\pm$ 0.20	91.30 $\pm$ 0.27	92.33 $\pm$ 0.14	92.40 $\pm$ 0.16	98.66 $\pm$ 0.11	94.82 $\pm$ 0.32	84.68 $\pm$ 0.26

4) Variants of STGVAD

All the variants discussed in this section operate on a common input format and produce outputs in a unified manner. For temporal modeling, we consider two sequence models: LSTM and Transformer. For spatial modeling, we employ three widely used GNN variants: GCN [20], GraphSAGE [16], and GAT [49].

- **GCN:** A spectral-based graph convolutional network that approximates localized graph filters by combining each node's features with a normalized aggregation of its neighbors' features. This smoothing operation allows information to propagate through the graph in a fixed, parameter-efficient manner.

- **GraphSAGE:** An inductive framework that samples a fixed-size set of neighbors for each node and applies a chosen aggregation function (e.g., mean, LSTM, pooling) to combine their features. By learning these aggregator functions, GraphSAGE can generate embeddings for previously unseen nodes or graphs.
- **GAT:** A graph attention network that replaces fixed-weight neighbor aggregation with learnable attention coefficients. For each node, self-attention is computed over its neighbors to adaptively weight their contributions before summing, enabling the model to focus on the most relevant context.

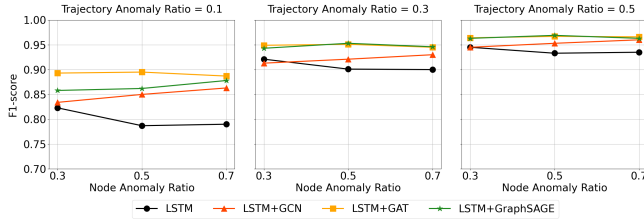


FIGURE 4: F_1 trends for **LSTM**-based models as a function of the *node-level* anomaly ratio (r_{node}) under three trajectory-level anomaly ratio (r_{traj}) settings

5) Hyperparameter Settings

We comprehensively evaluated nine model architectures (LSTM, Transformer, TrAISformer, and their GNN variants with GAT, GCN, and GraphSAGE) across nine anomaly injection scenarios. The training process utilized a learning rate of 0.01 to achieve optimal convergence while maintaining stability during optimization. A batch size of 4096 was used to ensure efficient processing. For model architectures, time-series models used 32 hidden dimensions, while GNN components in our hybrid architectures used 16 hidden dimensions, creating a hierarchical feature extraction pipeline where temporal features are first encoded and then refined through spatial modeling before final anomaly classification. All models were trained for up to 100 epochs with early stopping implemented using a patience of 20 epochs to prevent overfitting. Five different random seeds were used to ensure statistical significance of performance comparisons across all experimental configurations.

6) Evaluation Metrics

Given the binary nature of anomaly detection and the severe class imbalance in AIS trajectory data, conventional metrics such as accuracy may lead to misleading conclusions. We therefore report a comprehensive set of evaluation metrics that capture both majority class and minority class behaviors.

Let TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, respectively.

- **Precision:**

$$\text{Precision} = \frac{TP}{TP + FP}$$

Quantifies the reliability of anomaly predictions. High precision indicates fewer false alarms, which is crucial in maritime monitoring where unnecessary interventions are costly.

- **Recall:**

$$\text{Recall} = \frac{TP}{TP + FN}$$

Measures the ability to detect actual anomalies. High recall ensures safety by minimizing undetected abnormal trajectories (false negatives). In maritime monitoring, missing anomalous behavior can have severe consequences, making recall a critical safety metric.

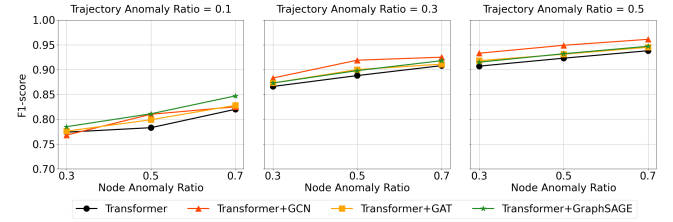


FIGURE 5: F_1 trends for **Transformer**-based models as a function of the *node-level* anomaly ratio (r_{node}) under three trajectory-level anomaly ratio (r_{traj}) settings

- **F_1 Score:**

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

The harmonic mean of precision and recall, balancing false positives and false negatives.

- **Accuracy:**

$$\text{ACC} = \frac{TP + TN}{TP + TN + FP + FN}$$

Measures the overall correctness of predictions. While intuitive, it tends to be biased toward the majority (normal) class under imbalance.

- **PR-AUC (Precision-Recall Area Under Curve):**

$$\text{PR-AUC} = \int_0^1 \text{Precision}(r) dr$$

Evaluates model performance across all decision thresholds by computing the area under the precision-recall curve. PR-AUC is particularly suitable for imbalanced datasets as it focuses on the minority (anomaly) class performance, unlike ROC-AUC which can be overly optimistic.

- **G-mean (Geometric Mean):**

$$\text{G-mean} = \sqrt{\frac{TP}{TP + FN} \times \frac{TN}{TN + FP}}$$

Balances sensitivity (recall) and specificity by taking their geometric mean. G-mean provides a single metric that accounts for performance on both classes, making it robust to class imbalance and ensuring neither class is ignored.

- **Matthews Correlation Coefficient (MCC) [7]:**

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

A correlation-based measure that incorporates all four elements of the confusion matrix. Unlike accuracy or F_1 , MCC is robust under severe class imbalance. Prior work highlights MCC as a more informative and reliable measure for binary classification, making it especially relevant for anomaly detection.

By jointly reporting these metrics, we ensure that the evaluation reflects not only overall classification accuracy but also

the reliability and robustness of anomaly detection under class imbalance. Furthermore, by leveraging these diverse metrics, we validate the robustness of our proposed framework across multiple anomaly injection scenarios.

B. RESULTS AND DISCUSSION

We conduct our experiments across varying anomaly trajectory ratios ($r_{traj} \in \{0.1, 0.3, 0.5\}$), with the anomaly node ratio fixed at $r_{node} = 0.5$. The reported values are the mean values after iterating each experiment by 5 times, and are scaled by 100.

The experimental results clearly demonstrate that STGVAD surpasses both the general time-series baselines (LSTM and Transformer) as well as TrAISformer, a recent state-of-the-art method designed for vessel trajectory detection. As summarized in Table 1, STGVAD consistently outperforms all purely temporal models across all trajectory anomaly ratio settings. The performance gap is particularly pronounced under low trajectory anomaly ratio conditions ($r_{traj} = 0.1$), which represent the most challenging and realistic anomaly detection scenarios where abnormal trajectories are rare.

Notably, the addition of GAT to the LSTM backbone yields substantial performance gains, particularly under low trajectory anomaly ratios ($r_{traj} = 0.1$), achieving improvements of up to +17.2 in MCC compared to the baseline. Transformer backbone also benefits significantly from spatial integration, especially with GCN, demonstrating consistent enhancements across all evaluated metrics, including a notable MCC gain of +2.9 in stress conditions ($r_{traj} = 0.5$).

Furthermore, Figures 4 and 5 underscore the robustness of the integrated spatial-temporal models across diverse node-level and trajectory-level anomaly ratios. Realistic low anomaly scenarios represent situations where anomalies are rare, making the detection task particularly challenging. Specifically, the LSTM-based models with GAT integration achieve accuracy improvements of up to +1.8, recall gains of +8.8, and MCC enhancements of +17.2, clearly illustrating the effectiveness of explicitly modeling node relationships in these challenging, low anomaly conditions. This effectiveness arises because LSTM inherently has fewer parameters and lacks a built-in attention mechanism, thereby limiting their ability to capture complex relationships among nodes. Incorporating GAT, which explicitly attends to relationships between nodes, significantly enhances the ability of the model to capture nuanced spatial correlations that would otherwise remain hidden.

C. ABLATION STUDY

We conduct comprehensive ablation studies to analyze two critical aspects of our approach: computational efficiency and the impact of temporal window size on performance.

1) Computational Efficiency Analysis

The proposed approach exhibits noteworthy computational efficiency while achieving substantial performance enhancements. As demonstrated in Table 2, the integration of diverse

TABLE 2: Training and inference time per epoch (seconds) on the test dataset.

Method	Training Time (s)	Inference Time (s)
LSTM	7.61 $\pm$ 0.29	1.37 $\pm$ 0.07
STGVAD (Ours)		
LSTM + GCN	7.94 $\pm$ 0.28	1.39 $\pm$ 0.07
LSTM + SAGE	7.81 $\pm$ 0.23	1.38 $\pm$ 0.06
LSTM + GAT	7.94 $\pm$ 0.24	1.44 $\pm$ 0.05
Transformer	8.89 $\pm$ 0.42	1.40 $\pm$ 0.10
STGVAD (Ours)		
Transformer + GCN	9.26 $\pm$ 0.46	1.73 $\pm$ 0.15
Transformer + SAGE	9.13 $\pm$ 0.42	1.46 $\pm$ 0.11
Transformer + GAT	8.90 $\pm$ 0.41	1.44 $\pm$ 0.11
TrAISformer	19.81 $\pm$ 0.31	1.84 $\pm$ 0.18

TABLE 3: Impact of temporal window size (w) on anomaly detection performance.

Metric	Window Size (w)			
	10	20	30	40
F1	89.48 $\pm$ 0.68	93.66 $\pm$ 0.45	93.77 $\pm$ 1.84	93.41 $\pm$ 3.60
MCC	88.39 $\pm$ 0.73	92.99 $\pm$ 0.49	93.13 $\pm$ 1.99	92.72 $\pm$ 3.96
Training Time (s)	7.82 $\pm$ 0.42	8.76 $\pm$ 0.13	10.95 $\pm$ 0.34	13.44 $\pm$ 0.65

GNN variants with LSTM backbones introduces a negligible computational overhead, amounting to a mere 0.20s to 0.33s augmentation in training time per epoch (2.6% to 4.3% increase) and 0.01s to 0.07s in inference time (0.7% to 5.1% increase). For Transformer backbones, the overhead remains modest, with training time increases ranging from 0.01 seconds to 0.37 seconds (0.1% to 4.2%), and inference time increases ranging from 0.04 seconds to 0.33 seconds (2.9% to 23.6%). It is imperative to note that this marginal computational cost yields substantial performance gains, resulting in optimal cost-effectiveness ratios. Among the GNN variants, SAGE consistently exhibits the lowest computational overhead while maintaining strong performance improvements. While LSTM baselines persist in their role as the most expeditious models, our LSTM-GNN variants have been shown to achieve superior accuracy with minimal additional computational expense. In contrast, TrAISformer requires a substantially greater amount of resources, with a training time that is approximately $2.2\times$ to $2.5\times$ longer and an inference time that is 25% to 34% slower, without demonstrating proportional performance benefits. This analysis demonstrates that the proposed method achieves an optimal trade-off between enhanced modeling capability and computational efficiency, providing superior performance-per-computation ratios that render it highly suitable for resource-conscious real-world deployments.

2) Window Size Sensitivity Analysis

We further conducted a sensitivity analysis of the time window size (w) under the Realistic-Low setting ($r_{traj} = 0.1$),

which is considered the most representative of real-world anomaly distributions. As shown in Table 3, increasing w from 10 to 30 consistently improved the performance across all metrics (F_1 , and MCC). However, when w was further increased to 40, the performance slightly degraded. We attribute this decline to the reduced absolute number of clusters formed during data partitioning, which limits the model's ability to capture fine-grained temporal patterns. Notably, while the performance gain from $w = 10$ to $w = 30$ shows a 4.8% F_1 improvement, this comes at a 40% increase in computational cost, indicating diminishing returns. For our main experiments, we selected $w = 10$ as it provides an optimal balance between detection accuracy and computational efficiency. Both training and inference costs increased almost linearly with larger w , reflecting the inherent trade-off between accuracy and computational efficiency.

V. CONCLUSION

In this paper, we propose STGVAD, a novel GNN-based framework for vessel behavior anomaly detection which introduces a new representation of maritime dynamics by modeling each timestamped vessel position as a node within a multi-ship trajectory graph. To leverage the existing ST-GNN's potential in spatio-temporal modeling, we retain the separation of temporal and spatial modules, while fundamentally redefining the underlying graph construction to accommodate the unstructured and anchorless nature of maritime environments. By incorporating spatial proximity via the OPTICS algorithm and temporal transitions into a single graph, our method captures both inter-vessel interactions and behavioral evolution over time. To rigorously assess anomaly-detection performance, we also generated controlled yet realistic anomalous trajectories on top of a real AIS dataset, enabling us to compare against traditional time-series models that do not exploit ship-to-ship interactions. Results show that our framework consistently outperforms traditional time-series models even in scarce anomaly settings.

In future work, we plan to extend our framework to an inductive setting, enabling generalization to unseen vessels and routes which is an essential capability for real-world deployment. We will also explore streaming graph construction methods to incrementally update spatial proximity as new AIS data arrives, avoiding full graph reconstruction. Additionally, we aim to shift from trajectory-level forecasting to fine-grained anomaly detection at the node level, which is more aligned with real-time maritime applications such as collision avoidance and compliance monitoring. Beyond AIS trajectories, we also envision extending our framework to incorporate data from other sensing modalities, such as sonar [56] and radar [15], by adapting the feature extraction and graph construction processes while retaining the core spatio-temporal modeling capability.

REFERENCES

- [1] E. M. S. Agency. Automated behaviour monitoring (abm) system. <https://www.emsa.europa.eu/we-do/digitalisation/86-combined-maritime-data/5177-abm.html>, 2018.
- [2] E. Akyuz. A hybrid accident analysis method to assess potential navigational contingencies: The case of ship grounding. *Safety science*, 79:268–276, 2015.
- [3] M. Ankerst, M. M. Breunig, H.-P. Kriegel, and J. Sander. Optics: Ordering points to identify the clustering structure. *ACM Sigmod record*, 28(2):49–60, 1999.
- [4] M. Anneken, Y. Fischer, and J. Beyerer. Evaluation and comparison of anomaly detection algorithms in annotated datasets from the maritime domain. In *2015 SAI Intelligent Systems Conference (IntelliSys)*, pages 169–178. IEEE, 2015.
- [5] V. Chandola, A. Banerjee, and V. Kumar. Anomaly detection: A survey. *ACM computing surveys (CSUR)*, 41(3):1–58, 2009.
- [6] X. Cheng, J. Wang, X. Chen, and F. Zhang. Attention-enhanced and integrated deep learning approach for fishing vessel classification based on multiple features. *Scientific Reports*, 15(1):8642, 2025.
- [7] D. Chicco and G. Jurman. The advantages of the matthews correlation coefficient (mcc) over f1 score and accuracy in binary classification evaluation. *BMC genomics*, 21(1):6, 2020.
- [8] J. Choi, H. Choi, J. Hwang, and N. Park. Graph neural controlled differential equations for traffic forecasting. In *Proceedings of the AAAI conference on artificial intelligence*, volume 36, pages 6367–6374, 2022.
- [9] M. Constapel, P. Koch, and H.-C. Burmeister. On the implementation of a rule-based system to perform assessment of colregs onboard maritime autonomous surface ships. In *Journal of Physics: Conference Series*, volume 2311, page 012033. IOP Publishing, 2022.
- [10] Y. Djenouri, A. Belhadi, D. Djenouri, G. Srivastava, and J. C.-W. Lin. Intelligent deep fusion network for anomaly identification in maritime transportation systems. *IEEE Transactions on Intelligent Transportation Systems*, 24(2):2392–2400, 2022.
- [11] E. d'Afflilio, P. Braca, and P. Willett. Malicious ais spoofing and abnormal stealth deviations: A comprehensive statistical framework for maritime anomaly detection. *IEEE Transactions on Aerospace and Electronic Systems*, 57(4):2093–2108, 2021.
- [12] L. Eljabu, M. Etemad, and S. Matwin. Anomaly detection in maritime domain based on spatio-temporal analysis of ais data using graph neural networks. In *2021 5th International Conference on Vision, Image and Signal Processing (ICVISIP)*, pages 142–147. IEEE, 2021.
- [13] P. Fu, H. Wang, K. Liu, X. Hu, and H. Zhang. Finding abnormal vessel trajectories using feature learning. *IEEE Access*, 5:7898–7909, 2017.
- [14] S. Guo, Y. Lin, N. Feng, C. Song, and H. Wan. Attention based spatial-temporal graph convolutional networks for traffic flow forecasting. In *Proceedings of the AAAI conference on artificial intelligence*, volume 33, pages 922–929, 2019.
- [15] Z. Guo, L. Wang, Z. Liu, Z. Fu, N. Li, and X. Zhang. Adaptive barrage jamming against sar based on prior information and scene segmentation. *Remote Sensing*, 17(7), 2025.
- [16] W. Hamilton, Z. Ying, and J. Leskovec. Inductive representation learning on large graphs. *Advances in neural information processing systems*, 30, 2017.
- [17] X. Han, C. Armenakis, and M. Jadidi. Modeling vessel behaviours by clustering ais data using optimized dbscan. *Sustainability*, 13(15):8162, 2021.
- [18] S. Hochreiter and J. Schmidhuber. Long short-term memory. *Neural Comput.*, 9(8):1735–1780, Nov. 1997.
- [19] Y. Jin, K. Chen, and Q. Yang. Transferable graph structure learning for graph-based traffic forecasting across cities. In *Proceedings of the 29th ACM SIGKDD conference on knowledge discovery and data mining*, pages 1032–1043, 2023.
- [20] T. N. Kipf and M. Welling. Semi-supervised classification with graph convolutional networks. *arXiv preprint arXiv:1609.02907*, 2016.
- [21] L. Klein, A. Foong, T. Fjelde, B. Mlodozieniec, M. Brockschmidt, S. Nowozin, F. Noé, and R. Tomioka. Timewarp: Transferable acceleration of molecular dynamics by learning time-coarsened dynamics. *Advances in Neural Information Processing Systems*, 36:52863–52883, 2023.
- [22] P.-R. Lei. Mining maritime traffic conflict trajectories from a massive ais data. *Knowledge and information systems*, 62(1):259–285, 2020.
- [23] H. Li, H. Jiao, and Z. Yang. Ship trajectory prediction based on machine learning and deep learning: A systematic review and methods analysis. *Engineering Applications of Artificial Intelligence*, 126:107062, 2023.
- [24] Y. Li, R. Yu, C. Shahabi, and Y. Liu. Diffusion convolutional recurrent neural network: Data-driven traffic forecasting. *arXiv preprint arXiv:1707.01926*, 2017.

- [25] N. Lim, B. Hooi, S.-K. Ng, X. Wang, Y. L. Goh, R. Weng, and J. Varadarajan. Stp-udgat: Spatial-temporal-preference user dimensional graph attention network for next poi recommendation. In *Proceedings of the 29th ACM International conference on information & knowledge management*, pages 845–854, 2020.
- [26] B. Liu, E. N. de Souza, C. Hilliard, and S. Matwin. Ship movement anomaly detection using specialized distance measures. In *2015 18th International Conference on Information Fusion (Fusion)*, pages 1113–1120. IEEE, 2015.
- [27] H. Liu, Z. Jia, B. Li, Y. Liu, and Z. Qi. The model of vessel trajectory abnormal behavior detection based on graph attention prediction and reconstruction network. *Ocean Engineering*, 290:116316, 2023.
- [28] R. W. Liu, M. Liang, J. Nie, W. Y. B. Lim, Y. Zhang, and M. Guizani. Deep learning-powered vessel trajectory prediction for improving smart traffic services in maritime internet of things. *IEEE Transactions on Network Science and Engineering*, 9(5):3080–3094, 2022.
- [29] K.-D. Luong and A. K. Singh. Fragment-based pretraining and finetuning on molecular graphs. *Advances in Neural Information Processing Systems*, 36:17584–17601, 2023.
- [30] C. Maganaris, E. Protopapadakis, and N. Doulamis. Outlier detection in maritime environments using ais data and deep recurrent architectures. In *Proceedings of the 17th International Conference on Pervasive Technologies Related to Assistive Environments*, pages 420–427, 2024.
- [31] T. Mantecón, D. Casals, J. J. Navarro-Corcuera, C. R. del Blanco, and F. Jaureguizar. Deep learning to enhance maritime situation awareness. In *2019 20th International Radar Symposium (IRS)*, pages 1–8. IEEE, 2019.
- [32] M. Masek, C. P. Lam, T. Rybicki, J. Snell, D. Wheat, L. Kelly, D. Glassborow, and C. Smith-Gander. The open maritime traffic analysis dataset. In *24th International Congress on Modelling and Simulation (MODSIM)*, pages 948–954, 2021.
- [33] L. May Petry, A. Soares, V. Bogorny, B. Brandoli, and S. Matwin. Challenges in vessel behavior and anomaly detection: From classical machine learning to deep learning. In *Advances in Artificial Intelligence: 33rd Canadian Conference on Artificial Intelligence, Canadian AI 2020, Ottawa, ON, Canada, May 13–15, 2020, Proceedings 33*, pages 401–407. Springer, 2020.
- [34] A. Mohamed, K. Qian, M. Elhoseiny, and C. Claudel. Social-stgcnn: A social spatio-temporal graph convolutional neural network for human trajectory prediction. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 14424–14432, 2020.
- [35] D. Nguyen and R. Fablet. A transformer network with sparse augmented data representation and cross entropy loss for ais-based vessel trajectory prediction. *IEEE Access*, 12:21596–21609, 2024.
- [36] D. Nguyen, R. Vadaine, G. Hajduch, R. Garelo, and R. Fablet. A multi-task deep learning architecture for maritime surveillance using ais data streams. In *2018 IEEE 5th International Conference on Data Science and Advanced Analytics (DSAA)*, pages 331–340. IEEE, 2018.
- [37] D. Nguyen, R. Vadaine, G. Hajduch, R. Garelo, and R. Fablet. Geo-tracknet—a maritime anomaly detector using probabilistic neural network representation of ais tracks and a contrario detection. *IEEE Transactions on Intelligent Transportation Systems*, 23(6):5655–5667, 2021.
- [38] I. Obradović, M. Miličević, and K. Žubrinić. Machine learning approaches to maritime anomaly detection. *Naše more: znanstveni časopis za more i pomorstvo*, 61(5-6):96–101, 2014.
- [39] G. Pallotta, M. Vespe, and K. Bryan. Vessel pattern knowledge discovery from ais data: A framework for anomaly detection and route prediction. *Entropy*, 15(6):2218–2245, 2013.
- [40] K. Patroumpas, E. Alevizos, A. Artikis, M. Voudas, N. Pelekis, and Y. Theodoridis. Online event recognition from moving vessel trajectories. *GeoInformatica*, 21:389–427, 2017.
- [41] Y. Qi, J. Yang, A. Bai, and J. Liu. A method for real-time detection of vessel abnormal behavior based on cnn-lstm. *Expert Systems with Applications*, page 128303, 2025.
- [42] M. Riveiro, G. Falkman, and T. Ziemke. Improving maritime anomaly detection and situation awareness through interactive visualization. In *2008 11th International Conference on Information Fusion*, pages 1–8. IEEE, 2008.
- [43] H. Rong, A. Teixeira, and C. G. Soares. A framework for ship abnormal behaviour detection and classification using ais data. *Reliability Engineering & System Safety*, 247:110105, 2024.
- [44] J. Roy. Rule-based expert system for maritime anomaly detection. In *Sensors, and Command, Control, Communications, and Intelligence (C3I) Technologies for Homeland Security and Homeland Defense IX*, volume 7666, pages 597–608. SPIE, 2010.
- [45] H. Y. Shahir, U. Glässer, N. Nalbandyan, and H. Wehn. Maritime situation analysis: A multi-vessel interaction and anomaly detection framework. In *2014 IEEE Joint Intelligence and Security Informatics Conference*, pages 192–199. IEEE, 2014.
- [46] H. Y. Shahir, U. Glasser, A. Y. Shahir, and H. Wehn. Maritime situation analysis framework: Vessel interaction classification and anomaly detection. In *2015 IEEE International Conference on Big Data (Big Data)*, pages 1279–1289. Ieee, 2015.
- [47] I. Varlamis, K. Tserpes, and C. Sardanios. Detecting search and rescue missions from ais data. In *2018 IEEE 34th international conference on data engineering workshops (ICDEW)*, pages 60–65. IEEE, 2018.
- [48] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need. In *Proceedings of the 31st International Conference on Neural Information Processing Systems, NIPS’17*, page 6000–6010, Red Hook, NY, USA, 2017. Curran Associates Inc.
- [49] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio. Graph attention networks. *arXiv preprint arXiv:1710.10903*, 2017.
- [50] S. Wang, Y. Li, H. Xing, and Z. Zhang. Vessel trajectory prediction based on spatio-temporal graph convolutional network for complex and crowded sea areas. *Ocean Engineering*, 298:117232, 2024.
- [51] Y. Wang, Y. Wang, Z. Zhang, S. Yang, K. Zhao, and J. Liu. User: Unsupervised structural entropy-based robust graph neural network. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 10235–10243, 2023.
- [52] K. Wolsing, L. Roepert, J. Bauer, and K. Wehrle. Anomaly detection in maritime ais tracks: A review of recent approaches. *Journal of Marine Science and Engineering*, 10(1):112, 2022.
- [53] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang. Graph wavenet for deep spatial-temporal graph modeling. *arXiv preprint arXiv:1906.00121*, 2019.
- [54] B. Yu, H. Yin, and Z. Zhu. Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting. *arXiv preprint arXiv:1709.04875*, 2017.
- [55] Y. Zang, L. Deng, S. Sun, Y. Ai, D. Geng, and X. Kang. Abnormal behavior detection of vessels based on deep learning algorithm: case study. In *2021 6th International Conference on Transportation Information and Safety (ICTIS)*, pages 205–212. IEEE, 2021.
- [56] X. Zhang, P. Yang, and D. Cao. Synthetic aperture image enhancement with near-coinciding nonuniform sampling case. *Computers and Electrical Engineering*, 120:109818, 2024.
- [57] Y. Zhang, H. Gao, J. Pei, and H. Huang. Robust self-supervised structural graph neural network for social network prediction. In *Proceedings of the ACM Web Conference 2022*, pages 1352–1361, 2022.
- [58] Y. Zhang, Q. Jin, M. Liang, R. Ma, and R. W. Liu. Vessel behavior anomaly detection using graph attention network. In *International Conference on Neural Information Processing*, pages 291–304. Springer, 2023.
- [59] R. Zhen, Y. Jin, Q. Hu, Z. Shao, and N. Nikitakos. Maritime anomaly detection within coastal waters based on vessel trajectory clustering and naïve bayes classifier. *The Journal of Navigation*, 70(3):648–670, 2017.



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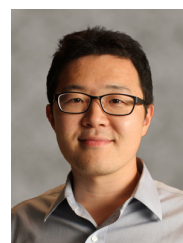


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